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Review

Equity and Spatial Justice Perspectives in Transportation

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Abstract: The concepts of equity and spatial justice may be relatively vague and therefore susceptible to different interpretations and metrics, leading to different evaluation perspectives. Thus, our central objective in this work is to explore the scientific production around these two concepts through the lens of transportation, focusing on their connections and interpretations. A mixed-approach analysis (network analysis and semi-systematic review) using Scopus, Web of Science, and Transportation Research International Documentation databases and VOSviewer software reveals that for transportation-related research, equity is more prominent than spatial justice. Accessibility emerges as a key concept for evaluating distributive issues in both contexts. The limited number of papers found may suggest either a gap in research or a mismatch between the terms within the field.

Keywords: spatial justice; equity; transportation; literature review

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1. Introduction

Although equity and spatial justice are concepts usually present in transportation policy, they may be relatively vague, and therefore susceptible to different interpretations and metrics that are ultimately reflected in project and policy evaluation processes. Therefore, a very relevant question concerns the definitions and subsequent interpretations and applicability of these concepts.

The discourse on justice is an increasingly prominent topic in the transportation literature. There are different approaches to justice (e.g., distributive, procedural, epistemic, restorative), but usually, the distributive justice approach is the most frequently used in evaluations of interventions in the transport system [1]. Moreover, the metrics and evaluation parameters used to assess transport intervention projects are directly influenced by the justice theory used as a background [2]. Justice is a central goal of planning, but there is a plurality of substantive concepts of justice, which different scholars defend and operationalize differently [3]. Therefore, equity and spatial justice are linked to the very concept of justice. From a wider perspective, justice (especially social justice) is intrinsically connected to how institutions exercise power over people and their rights; in other words, institutions are considered just when they do not draw arbitrary distinctions among individuals [4].

From a transportation-related perspective, justice entails the construction of a system offering individuals and social groups equal access to assets and opportunities to pursue their goals, in other words, removing barriers that prevent equity [5]. There is a relationship between spatial and social inequalities and the transportation system, resulting from the unequal distribution of resources and access to opportunities [6]. Along with this,

sociodemographic factors also influence mobility, as they end up restricting movement in many ways [7]. Because of its crucial role in shaping individuals' access to opportunities, fair public transportation must be ensured [8].

Our main objective in this work is to explore the transportation-related connections between the concepts of equity and spatial justice based on scientific production on the subject. With a focus on theoretical approaches, we report mainly on the operationalization (e.g., possible metrics adopted to measure and evaluate their parameters) and possible limitations of these concepts and their connections. To assess the amount and diversity of scientific knowledge produced on each of these topics separately, and with the aim of exploring the existing connection between the two topics, our methodology was divided into two parts: First, we performed a network and bibliometric analysis of the body of literature available in Scopus and Web of Science (WoS), using the bibliometric analysis software VOSviewer (version 1.6.20) [9], which allowed us to go further in the articulation of the sources of information contained in each article, unveiling chronological and thematic associations that would otherwise be difficult to find. Second, we conducted a semi-systematic review of the literature to identify theoretical approaches, operationalization, and gaps.

However, our research findings here do not represent an endpoint in searching for a suitable technique to understand spatial justice, equity theories, and transportation interactions. Thus, the findings are intended to support current methodological approaches that can provide new perspectives to policymakers on how to respond to the spatial and social disadvantages of some territories by means of mobility and transportation policies.

2. Main Concepts: A Brief

The most frequently used transportation-related justice approach is based on distributive justice, which concerns how benefits and burdens are distributed in society [10]. However, the concept of justice used as a background can significantly affect the evaluation [1,2,11]. Therefore, to better explore the concepts of spatial justice and equity, we must first turn to the very concept of justice itself, as it is on this definition that the other two concepts will base their premises and evaluation mechanisms.

For instance, utilitarianism is the most widely used theory of justice in assessments of transportation systems, usually materialized through global cost–benefit analyses. This approach is interested in maximizing the goods or service, regardless of how the transportation outcomes are distributed among socioeconomic groups [12,13]. On the other hand, Sen's Capability Approach (CA) is mainly concerned with promoting basic capability equality by ensuring a minimum level of basic capabilities for all [14]. In the CA, an emphasis on mobility and sufficient capabilities [14] provides a guiding point for transportation system distribution.

From another perspective, the concept of mobility justice pertains not only to expanding transportation infrastructures or accessibility, but also to exploring the cultural and social meanings surrounding the mobility infrastructures and modals [7]. It should address multiple scales of mobility and multiple approaches to justice in different areas (e.g., spatial and social). Also, justice in mobility can be evaluated at different scales, ranging from micro (interpersonal relations) to meso (issues related to urban transportation justice and "right to the city") and macro (borders and global resource flows) levels [7]. Transport justice [15] and mobility justice [7] are somehow complementary theories differing in focus. Transport justice emphasizes the identification of disparities in distributional effects. Mobility justice concerns overturning marginalization and disadvantages through intentionally including the excluded groups in decision-making and eliminating unfair privilege. When combined, justice in transportation aims to change the social system of mobility, considering that

mobility is a matter of many networked social relations and material processes at multiple scales [16].

Understanding the concept of distributive justice from a philosophical perspective allows us to combine quantitative and descriptive approaches [1], and can help in linking it with other concepts, such as participatory planning and spatial justice [1,17]. The debate on distributive justice focuses on the fundamental principle that should guide the allocation of goods [1,18–20]. Distributive justice is concerned with the provision of a "fair share" of goods, benefits, resources, and burdens to every member of society.

However, there are other types of justice approaches (e.g., epistemic, procedural, and restorative). The notion of the epistemic inclusion approach is based on the Schlosberg [21] framework of justice, which sees justice as an inseparable interaction between three components: distribution, recognition, and participation. The distribution dimension focuses on equity in the distribution of outcomes; the recognition dimension is a critical aspect of the framework, emphasizing the importance of true recognition of diversity (e.g., types of knowledge and ways of living); and the participation dimension focuses on the ability of individuals and communities to actively engage in a fair decision-making processes, requiring empowerment and democratic inclusion of all social groups, especially the marginalized and vulnerable. The recognition of diversity is also an integral part of the procedural justice approach [22].

Therefore, procedural justice focuses on issues related to ensuring "fair treatment" through fair decision-making and implementation processes. Ideally, decision-makers should be neutral, and groups affected by the decision should have representation and a voice in the process [23]. The restorative justice concept is concerned with repairing harm caused to certain groups or communities, and aims to prevent this harm from continuing to happen or prevent similar harm from occurring in the future. From a transportation perspective, a specific application of restorative justice is the "restorative equalization standard", which consists of providing historically disadvantaged communities with a disproportionate share of benefits to mitigate prior inequities [24]. It is important to emphasize that the application of justice theories to transportation-related problems presented here provides only a brief overview of the theme (for more information, see [1,15,24–29]).

Spatial justice is a relatively recent concept [17], and has been examined from different perspectives. The concept of spatial justice originates from the conceptualization of social justice in space, denoting a just distribution of available resources within and across geographical space [30]. The pursuit of spatial justice can be achieved if the organization of space and the allocation of resources across space respect the principles of equity and respect for human rights [31]. However, spatial forms of injustice can be linked to structural inequalities of different kinds [32]. Thus, work on spatial justice highlights the need to examine the pluralities of understanding the justice concept itself. The relationship of influence between spatiality and society depends on the theoretical perspective of spatial justice adopted. Thus, spatiality can be seen as a derivative of society, or as an active influencing factor on injustices [33–35].

Therefore, transport justice, mobility justice, and spatial justice seem to agree on the complexity of the dynamics involved in transportation systems, and on the fact that transportation should be understood as a resource. Moreover, transportation resources impact physical, human, and social capital [29]. This plural way of thinking about justice raises some relevant questions about the measurement of variation in justice and equity.

Equity refers to fairness in the distribution of impacts (benefits and costs) [11]. More specifically, transport equity refers to the fair distribution of transportation outcomes across spatial or sociodemographic groups [36]. However, building a robust measure that allows for a sound assessment can pose a challenge, as the concept itself involves several

parameters. There is a wide range of impacts and ways of measuring them. Also, there are several possible ways of grouping people for equity analysis. Both the adopted definition and equity measurement could significantly affect analysis results [11]. This means that the relationship between the concepts of equity and justice implies that adopting different definitions of justice will impact how equity is perceived, measured, and evaluated. As a result, assessing transport equity impacts can lead to conflicting results [11]. For instance, one perspective advances that a fair transportation system provides a sufficient level of accessibility to all under most circumstances [15]. This perspective has important implications for equity in transportation provision.

3. Methods

Snyder [37] maintains that regardless of the approach, every literature review should be divided into 4 phases: (I) designing the review, (II) conducting the review, (III) analysis, and (IV) writing up the review. Our study aims to understand the connections between the concepts of "spatial justice" and "equity", and how these concepts have been explored in relation to transportation. It is worth mentioning that despite equity and spatial justice being broader concepts, here, they will be explored from a transportation perspective. In this context, it is necessary to understand how the concepts have been addressed in the transportation literature produced so far.

The research took a mixed approach with two distinct stages: first, we carried out a bibliometric and network analysis of the literature (for each topic separately, and then all together); and second, we explored the selected articles through a semi-systematic (or integrative) approach to understand how the concepts of equity and spatial justice are addressed in the transportation-related literature, and drew our conclusions based on this.

3.1. Bibliometric and Network Analysis Database

Using the Scopus and Web of Science (WoS) databases, we conducted a search for titles, abstracts, and keywords (or "Topic" in WoS) using three keywords: equity, spatial justice, and transportation. In addition, the search code was adapted to include plurals and minor variations in the main terms (Table 1). Briefly, search 1 associated the keywords "equity" and "transportation", and search 2 associated the keywords "spatial justice" and "transportation"; both searches aimed to reveal the scope of each term within the field of transportation. Searches 3A and 3B (conducted in Scopus and WoS, respectively) used "equity" and "spatial justice" as keywords to explore the connections between these two topics in the literature. Finally, searches 4A and 4B associated all three keywords (equity, spatial justice, and transportation) to define the scope of the literature that explores the three topics together.

Table 1. Search parameters: Scopus and WoS results.

Search	Keywords	Domain	Filters (Limited to)		Results
			Document Type	Language	
1	"Equity" "Transport"	Title–Abstract–Keywords	Articles	"English"	3047
2	"Spatial Justice" "Transport"	Title–Abstract–Keywords	Articles	"English"	30

Table 1. *Cont.*

Search	Keywords	Domain	Filters (Limited to)		Results
			Document Type	Language	
Scopus					
3A	"Equity" "Spatial Justice"	Title–Abstract–Keywords	Articles	"English"	64
4A	"Equity" "Spatial Justice" "Transport"	Title–Abstract–Keywords	Articles	"English"	7
Web of Science					
3B	"Equity" "Spatial Justice"	Topic	Articles	"English"	64
4B	"Equity" "Spatial Justice" "Transport"	Topic	Articles	"English"	8

3.2. Semi-Systematic Database

An integrative review can be defined as a method of research that critically reviews and synthesizes the representative literature in order to generate new perspectives on a topic [38]. Along the same lines, Garvey [39] argues that a semi-systematic review focuses on generating new, mainly qualitative, insights from the interdisciplinary literature, rather than fully capturing and quantitatively analyzing records. Therefore, the review design was based on three core research questions that were developed through a scoping study of the literature, identifying the main descriptive (question 1) and evaluative (questions 2 and 3) research gaps in discussions of equity and spatial justice:

I. Which equity and spatial justice approaches are considered in evaluations of transportation projects and policies?
II. Which metrics are used to appraise transportation projects and policies? Which variables are used? Which social groups are considered?
III. Which limitations appear most frequently in studies?

In addition, the review approach consisted of two phases: first, searching for papers, excluding irrelevant papers, and selecting the relevant literature; and second, analyzing the selected articles. The search was conducted in accordance with the PRISMA guidelines for systematic reviews [40] and with the integrative review guidelines [38], with both approaches adapted for the present study.

Figure 1 provides a flow diagram (adapted from Liberati [40]) for the selection of scientific papers for inclusion and exclusion from this review. The flowchart presents the first phase of the integrative review process. The process (Figure 1) began with the identification of the keywords "equity" and "spatial justice" in the scope of scientific production contained within two major academic databases, Scopus and WoS. A total of 168 scientific productions (95 from Scopus and 73 from WoS) were identified. Following this, we combined the two samples and applied four specific filters to the samples; that is, to be eligible, the productions had to be papers, written in English, in their final version, and available for reading (open access). After applying these filter criteria, 132 papers from the unified database were selected for the next step. Subsequently, we removed duplicate articles, resulting in a sample of 85 papers. As our objective was to evaluate the academic production on the topics of equity and spatial justice applied to transportation, the last part of the screening process involved selecting only articles that addressed the

transportation theme, narrowing the selection down to 13 papers. Additionally, we added 4 more articles, resulting from our search for the keywords (equity and spatial justice) together in the Transportation Research International Documentation (TRID) database. After all filtering and selection steps, the final sample (17 papers) was established for the semi-systematic study.

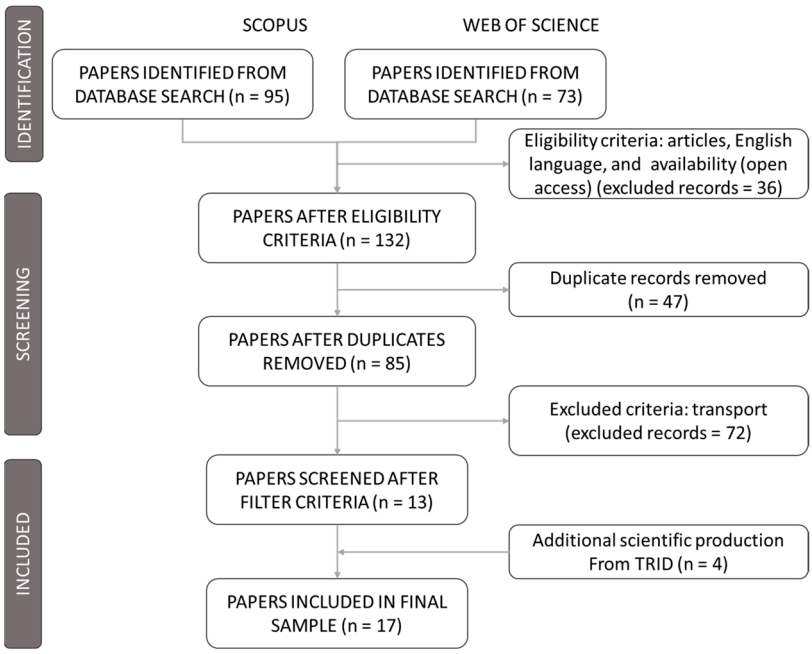

Figure 1. Overview of methodological approach adapted from PRISMA.

4. Results and Discussion

4.1. Bibliometric and Network Analysis

Three separate stages comprised the bibliometric analysis (Table 1): the first was more broadly oriented; the second concentrated on the link between the two key concepts; and the third investigated this inter-relationship from a transportation perspective. The first stage focused on the metrics of scientific production; the other two stages were more focused on understanding the scientific connections between terms. Based on the findings of the first stage (search 1 and 2), for each of the two concepts—spatial justice and equity—separately, under the filter "Transportation", it is possible to conclude that, except for a single paper published in 1965, the term "Equity" began to be studied in the 1970s, and has since maintained a steady growth rate of publications, reaching a peak of scientific production in 2021 and 2022. However, it is worth noting that for the two peak years, approximately 27% and 30% of the publications refer to COVID-19, respectively. In total, the search for the "Equity" term resulted in 3047 papers distributed across 112 countries or territories, with most of the publications (about 40%) from the United States of America (USA). Regarding the term "Spatial justice", the search findings reveal more modest values of scientific production; 30 articles were found in total, distributed among 25 different countries or territories, with the USA being the most representative (27% of the publications).

The second stage (3A and 3B) aimed to explore the relationship between equity and spatial justice. The search in Scopus (3A) identified 91 scientific publications, but 27 of these were not articles, and so were removed from the results. Thus, the remaining 64 publications represent the scientific literature that explores the equity and spatial justice concepts together. The search in WoS (3B) returned 73 scientific publications, with only 64 of them being published articles. Therefore, the results from Scopus and WoS were merged into a single database, which, after removing duplicate items, contained 81 scientific articles

published and written in English. Based on this database, it is possible to state that the scientific production exploring these two concepts is concentrated in the 21st century. From the 2010s onwards, it is possible to notice a growth in the number of articles, with the majority coming from the USA (20%). Articles linking equity and spatial justice are well distributed worldwide, among 46 countries or territories. The USA, China, and the United Kingdom have the highest scientific production; together, they are responsible for about half of all articles exploring the link between equity and spatial justice. Social sciences, environmental sciences, and engineering appear as the main areas of knowledge, with contributions of about 43%, 16%, and 7%, respectively.

The network visualization (Figure 2) shows that a wide variety of connections can be established between "equity" and "spatial justice" in different areas. The number of co-occurrences determines the size of each keyword. For instance, "sustainability", "justice", and "social justice", despite showing connections with both, have a stronger connection with spatial justice than with equity.

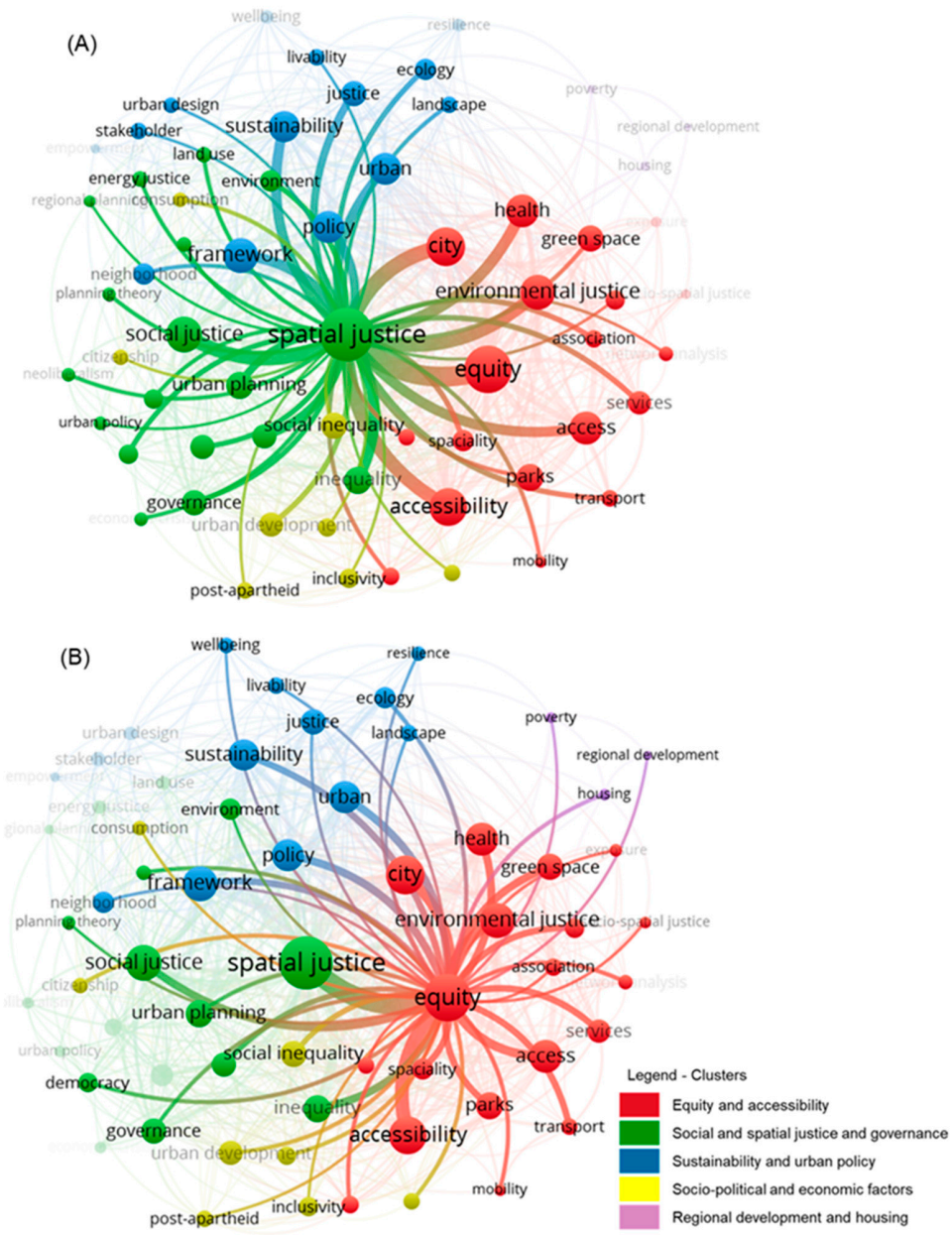

Figure 2. Co-occurrence network isolating (**A**) "spatial justice" and (**B**) "equity". Source: VOSviewer.

From the analysis of the network associations (Figure 2), it is possible to delimit five different thematic clusters based on the association strength between the keywords. The size of the nodes reflects the importance or frequency of a term, while the connections between them illustrate the relationships and co-occurrence of concepts in the literature. The green cluster highlights social and spatial justice and governance, exploring the role of urban policy and social inequality in shaping spatial justice from a more theoretical approach; the red cluster focuses on equity and accessibility, especially in transportation and mobility; the blue cluster connects urban sustainability with ecological and land use concerns; the yellow cluster investigates urban governance and historical injustices; and the purple cluster focuses on housing and regional development, emphasizing economic and social disparities, especially in discussions about environmental and social justice.

Additionally, when we isolate intermediate terms (Figure 3), it is possible to conclude that "justice" and "accessibility" have several dispersed connections, which could be an indication of the flexibility of their application. The term "accessibility" seems to be more prone to being related to transportation-related keywords (e.g., "transport", "mobility", "services", "inclusivity", and "disparities"). "Transport" has a higher level of connection with "equity" than with "spatial justice", and is connected with terms that refer to the distribution and evaluation of transportation in urban contexts (e.g., "city", "access", "accessibility", "services", and "mobility").

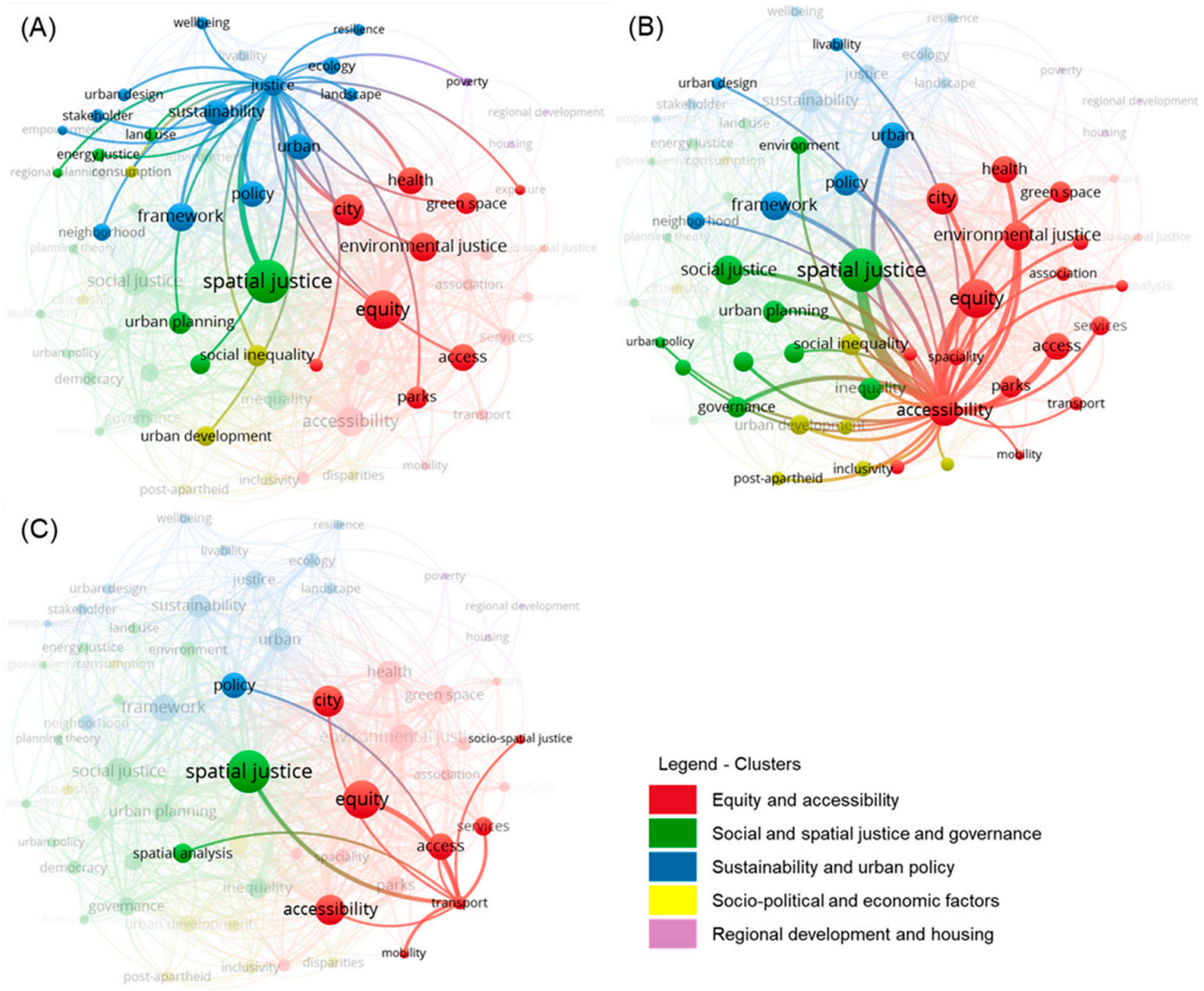

Figure 3. Co-occurrence network isolating (**A**) "justice", (**B**) "accessibility", and (**C**) "transport". Source: VOSviewer.

The final stage focused on exploring the links between equity and spatial justice from a transportation perspective, so we added the keyword "transport" as a delimiter in the search parameters. The search in Scopus (4A) revealed seven published scientific productions, and the search in WoS (4B) revealed eight articles published in English. Therefore, combining the Scopus and WoS results and eliminating duplicates, the resulting sample contained only nine articles. Due to the limited number of publications in our sample, we decided to add another four relevant publications. Therefore, the final database exploring the three concepts simultaneously consisted of 12 papers. The scientific production that explores the connection between equity, spatial justice, and transportation is quite current (starting in 2015), but still modest, with an average of two articles per year from 2022 onwards. Social science and environmental science are the main subject areas, accounting for 70% of the articles (respectively, 50% and 20%). Canada, the Netherlands, and South Africa are the countries that show the highest scientific output relating to the three keywords.

The analysis of the distribution of network connections allows us to see that "equity" is the broadest theme, allowing a greater diversity of connections. Another interesting insight is based on the isolated appearance of the term "urban transport", which emphasizes that these studies are focused on the urban context. However, the three main terms are broad, allowing them to be used in several different contexts. Due to the small number of publications, the number of associated keywords occurring simultaneously in at least two articles is quite small.

4.2. Semi-Systematic Analysis

Since equity and spatial justice can be quantified and understood from various perspectives, to answer the first research question, we decided to extrapolate the scope of the articles and work with the broader literature to better explain the origin and evolution of the concepts. The other two research questions are answered by focusing on the database obtained in the systematic search. Moreover, it is possible to see from the bibliometric analysis that the concepts of spatial justice and equity do not usually appear simultaneously in relation to transportation.

4.2.1. Concepts from a Transportation Perspective

There is a set of justice approaches that can potentially influence the assessment and the chosen metrics in this research field, as mentioned at the beginning of this study. Here, we focus on the concept of spatial justice, and relate it to the discussion of equity issues in transportation. Spatial justice is a theoretical concept that is rarely used to measure and evaluate transportation-related justice issues. One of the main lines of discussion in the field of spatial justice regards the meaning attributed to space and its relation to society [33]. However, spatial justice can be associated with issues of equity and accessibility.

The origin of the spatial justice concept goes back to the "right to the city" concept proposed by Lefebvre [41]. Later, Marcuse [42,43] and Soja [17] explored this concept through an urban studies approach. They advanced the theory of relationships between exclusion, space, and society.

Spatial injustices are always derived from broader social injustices, as the latter always have a spatial aspect [42]. Harvey [44] explores the dynamics between society and space, and their relationship with capitalism. Injustices are an integral part of capitalism and its neoliberal manifestation, and justice cannot be achieved without a structural transformation in the system [44]. Therefore, seeking other forms of solidarity and collectivity, rather than an exclusive search for profits, can be a common cause that involves different groups in the fight for the right to the city [43]. For these scholars, the main goal is social justice, and spatiality is cast as the geographical dimension of social processes [33].

From a different perspective, and following Lefebvre [41], Soja does not see spatial (in)justice as derivative or as a sub-item of a broader concept of "just city" [45]. Based on this premise, space is a social product, and understanding spatial phenomena depends on social and cultural textures [17]. The spatiality of (in)justice affects society and social life, and is itself shaped by social processes [17]. The transcription of human activity in space at any time in history and in any social context invariably leads to situations of social injustice and spatial exclusion (deriving from centralities and the hierarchy of places, accessibility, etc.) [17].

Spatial justice focuses on how resources are distributed across space, and how the spatiality of living influences, enhances, or confines people in their opportunities [35]. Thus, the patterns of (dis)advantages could be concentrated in some areas, which may cause or increase further processes of inequality and marginality [33]. Relating mobility and spatial justice perspectives, an increase in urban spatial justice should prioritize improvement of modes of transport mainly used by deprived individuals or people with disabilities [46]. The interaction between individuals, their activities, and movements continually reorganizes space, generating different levels of accessibility, based on the assumption that space consists of resources (which can be distributed) [46]. Therefore, one possible interpretation of the concept would be that spatial justice is a struggle for equity in social space [33]. From another perspective, scholars argue that the spatial justice concept is a derivative concept, where space is the context of the conceptualization of justice [4]. For instance, from a more empirical perspective, spatial justice issues can be explored within the transportation equity context by focusing on how transportation resources are allocated and the implications of this distribution on social equity [47].

Equity can be understood as the way of conceiving equality by treating groups that are at a disadvantage compared to others differently [48]. In fact, true equity implies that an individual or group may need to experience or receive something different (not equal) to facilitate fairness and access [5]. In transportation, equity concerns include, as their key components, the distribution of benefits and costs among members of society, along with the distributive principle (used to justify whether the distribution is "socially acceptable") [36].

Transportation-related equity aims to associate social and spatial factors in social welfare assessment through the concept of access to key activities [49]. For instance, this may involve minimizing disparities between less and more privileged groups by maximizing the average accessibility [26]. In theory, effective equitable transportation planning should help direct urban regions towards a more just transportation system [50]. However, transportation evaluations from an equity perspective can be difficult, because there are several definitions of equity, ways of categorizing people, impacts to consider, and ways to measure these impacts [11]. A transportation system is equitable if it provides a minimum level of accessibility for everyone under most circumstances [15] and disproportionately benefits less well-off groups [51].

Equity is often divided into horizontal and vertical categories. Horizontal equity deals with the equal distribution of an attribute among equal members of a population. Vertical equity involves equal distribution of an attribute among individuals and groups that differ by income, social status, or other applicable specificities in their evaluation [11,52]. Therefore, the focus on choice and mobility concerns is derived from Sen's theory of the Capability Approach [14], based on leveraging the concepts of capability and wellbeing to address transportation inequalities. Additionally, equity evaluations should address inequality issues from different perspectives, such as travel behavior and individual preferences, experiences, and capabilities [53].

Transport equity is mostly focused on the distribution of benefits and its impact on transportation-disadvantaged communities [15,36], based on a distributive justice approach [1,20]. After exploring the applicability of five different approaches to transportation-related justice and their limitations, Pereira [1] ultimately defines transportation-related justice as a moral and political ideal that relates to the distribution of benefits and burdens across society (distributive justice). In a procedural justice approach associated with a social justice perspective, an equity assessment should be able to capture structural biases in planning and funding practices that favor advantaged over disadvantaged groups, along with proposals that mitigate these issues [11].

The transport justice concept proposed by Karner [24] suggests adding a wider range of actors and concerns to the transport equity concept, especially in relation to issues regarding recognition (e.g., respect for the basic rights, needs, and values of involved or affected groups). In short, transportation justice describes a normative condition in which no person or group is disadvantaged by a lack of access to the opportunities they need to lead a meaningful and dignified life [24]. Thus, the concept approximates justice distribution issues to models of social change. However, the concept of transport justice is like the concept of spatial justice in that it focuses on a more theoretical approach to transportation issues, without direct guidelines for applicability and evaluation, which ultimately makes it challenging to apply these principles to assessments of transportation systems. In this sense, equity is more readily translated into assessments of transportation system initiatives, even though it is sensitive to the theoretical justice approach adopted [2,20]. The transport justice framework, besides its functionality as a guide for high-level transportation planning, can also be applicable at the design levels [51].

Finally, papers exploring transport justice, spatial justice, and equity highlight the roles of different actors in building and fostering urban mobility, and the effects that these processes have had on accessibility inequalities at various scales [19]. Although theoretical discussions around transport equity and justice are relevant, they frequently seem to be disconnected from empirical studies [18]. That said, the distributive justice approach is the most prominent and widely used in equity assessments, and the distribution of goods/services in a territory is also a focal point in spatial justice discussions. The distributive justice approach considers two primary elements: the distribution principle, which dictates the moral and socially acceptable precepts of distribution of the good or service; and the affected areas and/or populations [18,20]. Distributive justice approaches can embrace different principles of justice, and consequently, different ways of measuring goods and services, as well as different ways of classifying population groups [11].

4.2.2. Operationalization

It is certain that transportation services inevitably yield different levels of costs and benefits across an urban area, as well as different levels of accessibility. Transportation disadvantages directly impact the livelihood of communities and can contribute to social exclusion [28]. Despite the transportation literature exploring accessibility levels, mobility (travel behavior), and transportation externalities and resources as categories of benefits and burdens [54], examining accessibility and travel externalities has proven more productive for transport equity studies [20]. Social inclusion and exclusion statuses emerge from the interplay between income, access to job opportunities, housing availability, and basic access to goods and services [55].

Generally, the distinctions often made from an equity perspective are those between income strata and space. These distinctions could prevent the implementation of policies whereby low-income categories or poor regions "lose" and high-income categories or regions "win" [56]. Lewis [29] emphasizes that generic categorizations (e.g., horizontal or

vertical equity) leave too much room for interpretation; in contrast, using categories based on established concepts (e.g., space, time, or types of capital) are less prone to misuse.

Accessibility is intimately connected to transportation systems [57,58]; some authors maintain that the concept of accessibility ought to be central to comprehending transportation benefits and equity [1,15,20]. The notion of transportation accessibility (linking people and places) can be understood in terms of ability and freedom of choice, finding some support in the concept of the right to the city [41,44], in the spatial dimension, relating to moral concerns regarding the distribution of opportunities (goods/services); and in the concept of the Capability Approach [14], by setting a minimum accessibility threshold for people's ability to reach key activities.

The accessibility indicators for transport justice and equity analysis should capture opportunities that have an impact on people's wellbeing [59], namely basic access [11] or access to key activities [49,60]. There is a link between accessibility levels and socioeconomic status [61]. Therefore, individuals' accessibility is a reflection of the relationship between the transportation system, land use patterns, and individual attributes (e.g., income, gender, place of residence) [1,15].

Accessibility is a crucial factor in determining quality of life, as it reflects the potential for individuals to interact with and access various opportunities dispersed across space [28]. Geurs and Wee [62] maintain that accessibility indicators should consider four key components: land use, transport, temporal, and individual components. Together, the components measure how easily individuals can access opportunities. However, accessibility can be measured in different ways and from different ethical perspectives [1,56], and the choice of metric strongly shapes the conclusions [1,2,61].

Most commonly in the literature, accessibility is calculated using infrastructure-based measures, contour (or isochronic) measures, gravity measures, competition measures, utility-based measures, network measures, and time–space measures [63]. However, from an equity perspective, Di Ciommo and Shiftan [36] maintain that there are two types of measures for assessing accessibility to key activities: (I) isochronic measures (or cumulative opportunities), which consist of summing up the opportunities reachable within a given travel time, fixed cost, or travel distance; and (II) gravity measures, based on the concepts of attraction and impedance.

In most studies dealing with justice and equity, accessibility appears as a mechanism for comparing different realities and scenarios. Furthermore, "advantaged" and "disadvantaged" statuses are often established by socioeconomic characteristics (e.g., income, gender, race, etc.) and the jobs–households mismatch. It is possible to establish a relevant sociospatial link between people who are more prone to transportation disadvantages in terms of their location in the urban structure, and their spatial and temporal access to public transportation services [64,65]. Thus, transport justice scholars tend to focus on the concept of accessibility, especially from a distributive justice perspective [1,15,36,58,66].

Based on the literature, accessibility-based equity analysis is primarily conducted through two equity measures: the first is derived from economic indicators (e.g., Palma index, Lorenz curves, and the Gini index), and the second involves statistical indicators (e.g., Getis–Ord statistic). Typically, the Palma index [67] and the Gini index [68] are used in the transportation literature to assess disparities and compare the distribution of transportation-related costs and benefits, especially for accessibility levels. Initially, the Gini index and Lorenz curves were responsible for most studies' equity analyses. Since the Gini index considers the distribution among all groups, it is not sensitive to changes in the lower part of the distribution. As a response to this limitation, the Palma index began to be used, because it focuses only on extreme groups (the richest and the poorest) [8,69], and facilitates greater ease of interpretation and communication of results.

However, accessibility can be measured in many ways, and from different ethical perspectives [1,26,54,56]. For instance, Geurs and Wee [62] argue that a measure of accessibility should ideally consider four components—land use, transportation, individual, and temporal components. From our database, Nyamai and Schramm [46] argue that the accessibility framework and further assessment are composed of interactions between three mutually dependent dimensions—spatial, individual, and modal [46]. Therefore, the interplay among these three dimensions reinforces the concept of spatial justice, with mobility providing a metric to assess the distribution of resources.

As a general rule, studies focus on a distributive justice approach. For instance, Alizadeh [70] explores broadband network distribution through a spatial justice approach in Australia by associating and comparing socioeconomic indices (e.g., economic resources, education, and occupation) with the accessibility/remoteness index (ARIA). van Dijk [57] explores the equity effects of a toll charge on traffic diversion and work location accessibility, using income categories by transportation zones as a comparability factor. Pede and Staricco [71] explore the distribution of car-sharing services from a sociospatial justice perspective using the deprivation index (based on five indicators: education, housing condition, overcrowding, unemployment, and non-homeownership) and levels of service. Kim [72] uses multivariate regression models to evaluate the effect of low-income neighborhoods on the distribution of car-sharing supply and utilization. Qiao and Yeh [73] focus on the spatial justice approach and its link with accessibility distribution; they associate travel time-based regression accessibility with agent-based modeling to compare travel change scenarios.

From a different perspective, mobility-related issues of (in)justice extend beyond accessibility and transportation to uneven freedom of movement (such as the inner relations of race, gender, age, disability, and sexuality) [16,46,74]. Despite being concerned with broader issues of recognition and exclusion, mobility studies exploring the concept of accessibility focus not on the measures used, but on the groups investigated.

Karner [24] maintains that a combination of State- and society-centric strategies for public involvement in transportation planning can lead planners to establish new collaborative relationships and elevate the concerns of disadvantaged communities. However, most studies do not explore the issues of recognition and exclusion of groups from deliberative processes or other typologies of justice. Ramírez Saiz [75] highlights the relevance of addressing possible conflicts faced by individuals of different ages and abilities in the urban environment, through the lens of the 15 min city perspective.

Disadvantaged groups (often defined by income, literacy, age, or gender) are frequent and recurring subjects in the justice and equity literature for the assessment of disparities and inequalities. Individual experiences in transportation are shaped by the intersection of multiple socioeconomic identities, which, together, determine their privileged or disadvantaged status [49,76]. The overlap of various identities frequently creates additional difficulties for vulnerable populations, making members of many marginalized groups more likely to experience heightened exclusion or limited access to opportunities and services [8,77]. For instance, those from lower socioeconomic backgrounds may face greater barriers in basic access (e.g., education, healthcare, and employment), with these barriers being further exacerbated for individuals who are also part of minority ethnic or gender groups [28]. This intersectionality highlights how systemic inequalities are not experienced in isolation, but are instead shaped by interconnections between social identities [58].

According to Karner and Niemeier [77], target populations should be defined from a concentrated disadvantage perspective, as the various socioeconomic characteristics that place individuals at a disadvantage can overlap. Therefore, target populations are often identified using combinations of demographic factors (e.g., ethnic minority proportions, low

income, recent immigrants, single-parent families, etc.). A good alternative for categorizing populational groups is the target group position index (TPI), which is designed to analyze the situation of one or more subgroups within a population using two types of variables: a classifying (qualitative) and a ranking (quantitative) variable [58].

Finally, in transport justice and/or equity studies, travel time is the most widely used variable for empirical evaluations of travel behavior and mobility patterns, and is used to establish accessibility levels [18,46,51,57,64,73,75,78]. It is worth mentioning that in developed countries, the monetary cost of travel in relation to income has a smaller impact on accessibility levels than in developing and peripheral countries, where this impact ends up reducing the level of access to opportunities for a large portion of the population [78].

4.2.3. The Main Limitations Reported by the Studies

There is still a gap in the literature when it comes to tackling transport justice topics and empirical real-world data [18], especially in turning the assumptions of justice provided by the various theories into metrics that can be evaluated and compared. Thus, one of the recurring limitations in issues involving equity and spatial justice is the relationship between the concept of justice itself and its applicability to transportation.

The issue of distributive justice relates to how institutions and society itself shape social and economic inequality. Therefore, equity also refers to participation in transportation decision-making processes [20]. Bailey and Grossardt [79] point out that one of the main issues is the quality of the representation obtained in the participation process, and that those chosen for the process should be able to express their opinions on the projects. Social assessments that are extensively reliant only on census data may not fairly represent social diversity [70]. Usually, bottom-up strategies consider the majority or stereotyped groups, instead of those considered vulnerable. As the State is still involved in transportation planning and policies, societal actors still do not have a relatively more powerful position in the decision-making process. Ramírez Saiz [75] maintains that there is a divergence between acknowledging the need to consult with neighborhoods regarding urban decisions, and applying this in reality.

Another important limitation often mentioned concerns the quality of databases and the availability of sufficiently robust data for analysis [18,57,70,71]. Especially concerning patterns of travel behavior and levels of accessibility, regarding the latter, the use of zone-based accessibility measures can standardize people who experience completely different levels of accessibility [57,80].

5. Discussion

Our study employs a mixed approach (network analysis and integrative review) to analyze the scientific production on spatial justice, equity, and transportation together, revealing important distinctions in the application of the first two concepts. The results reveal that equity is more prominent in transportation studies, particularly from a distributive justice perspective, while spatial justice is more frequently associated with public policy and governance.

Accessibility emerges as a central concept and metric in equity and spatial justice evaluations. However, despite its importance, there is no universal metric for measuring and evaluating accessibility. From a transportation equity perspective, the distribution of accessibility levels should be a local matter based on the social relevance attributed to opportunities in a specific community [2,15]. However, a minimum level of access to essential activities must be ensured [15]. Accessibility indicators vary significantly depending on the justice parameters adopted. Travel time remains the most commonly used metric in evaluations. It is frequently associated with gravitational models, which

can capture the spatial dispersion of opportunities and scale them through an impedance factor, thereby making both the metric and the model the most commonly used and easily reportable outcomes.

However, Schlosberg [21] argues that a focus on distribution may not be sufficient for resolving justice issues. A promising alternative is to associate transportation with land use, housing, and institutional barriers [20]. Therefore, the direct and indirect relationships between accessibility, social exclusion, and equity effects should be explored through indicators that are sensitive to these effects [56]; this involves using more comprehensive measures of transportation costs, based on actual travel times and generalized travel costs, and associating them with composite social indicators. By incorporating these indicators, transportation planners can design more inclusive and equitable transportation systems that better meet the needs of individuals, particularly those from marginalized or minority groups.

Despite the discussion of justice gaining prominence in the transportation literature, there is still a gap between theoretical contributions, particularly regarding the conceptualization of the terms equity and spatial justice, and empirical measures capable of assessing this dimension of justice in transportation research. Additionally, research identifies a gap in the literature regarding the intersection between spatial justice and equity in transportation. Furthermore, while equity studies tend to focus on empirical assessments using accessibility as a measure of distributive justice, research on spatial justice leans towards theoretical approaches related to recognition and participation in decision-making processes. It is also worth noting that, despite being widely used, accessibility metrics remain context-dependent, presenting challenges for the establishment of universal standards in transportation justice. Finally, the review highlights the complexity of defining justice standards in transportation-related assessments, reinforcing the notion that space itself plays an active role in shaping social and territorial injustices. However, due to the small number of publications, these bibliometric results may not be sufficiently robust and representative.

From the semi-systematic results, it can be concluded that equity issues are more applicable and empirically transcribed, and are still often related to the distributive justice approach, while spatial justice studies are focused on theoretical approaches, especially on issues of recognition of both social and structural disadvantages. Furthermore, equity and spatial justice are broad concepts, and are closely linked to and dependent on the concept of justice itself. Thus, as there is no single, all-encompassing definition of justice, there is quite a variety of interpretations, connections, and applications.

Equity scholars end up exploring the distributive justice approach mainly by using accessibility as a measurement and comparison parameter. The accessibility indicator is often constructed using variables of travel time, travel distance (to the CBD and/or opportunities), and access (affordability, mobility needs), and evaluations often compare groups based on income, disability, age, gender, and other factors.

Spatial justice is mostly concerned with the spatiality of injustices, exploring the context of recognition of already embedded injustices. From a spatial justice perspective, transport justice involves the right to move around and access goods and services, and the capacity of individuals to transform spatial relations of movements, ultimately managing collective agency and freedom [16]. Spatial justice scholars usually focus on theoretical and procedural approaches, especially on issues related to decision-making processes (deliberative processes and public participation), generally considering disparities regarding gender, age, race, and sexuality, and issues related to recognition of inequalities (structural, historical, or merely distributive). However, it is worth noting that the concept of spatial justice also explores distributive issues of people and opportunities.

The transportation-related common ground is related to the distributive justice approach, but in both cases (equity and spatial justice), there is no ideal or universally accepted metric or systematic procedure to be applied. Due to this, the most prominent constraint regards translating the concept of justice (different justice approaches and different theories of justice led to different conclusions) into evaluation procedures and metrics that allow for comparability.

In general, scholars of transportation-related justice tend to focus on the concept and metric of accessibility, and assess the distribution of its levels across space or individuals. For this reason, accessibility plays a central role in transportation evaluations. In addition, a significant proportion of studies use low-income populations as a priority group and income levels as a classification variable. However, an individual's disadvantaged status is a summation of socioeconomic identities that overlap to create experiences of privilege or disadvantage in transportation. Vulnerable groups often face compounded barriers due to these intersecting identities, meaning that individuals who belong to multiple marginalized groups may experience heightened levels of discrimination and exclusion, or at least more limited access to opportunities and services.

6. Conclusions

This study demonstrates key conceptual and empirical distinctions between equity and spatial justice in transportation research. While equity is more often empirically measured (particularly through accessibility), spatial justice remains grounded in theoretical frameworks related to recognition and participatory governance.

One of the most important conclusions of this work is that the debate around justice and the definition of standards and limits results in different metrics and empirical evaluations of transportation systems and of projects to change the infrastructure of transportation systems. This is important, since establishing a threshold can be a relevant foundation for setting goals and public policies in the field of equity and justice [18]. However, most studies also show that metrics and their subsequent evaluation are context-dependent, which makes it difficult to create a generic parameter or indicator that fits all situations. Currently, in the literature, various indicators and metrics are used to empirically measure and evaluate disparities in a population based on the logic of equity or justice, but each of them ends up providing different descriptions of these disparities.

Concerning the small number of articles found on scientific dissemination platforms, two conclusions can be drawn: either that there is a gap in the literature when it comes to exploring the two themes together under the aegis of transportation, or that this combination may not be significant for current transportation issues. We tend to agree with the former, since space can be understood as an active actor in the formation, distribution, and aggravation of social, cultural, and territorial injustices. There is a limited number of publications that combine the three themes. Although it was not explored in this literature review, future studies aimed at exploring the possible reasons for this gap should focus on the transition from the concept of spatial justice to a focus on the territorial cohesion concept, particularly in the European context [33]. Another relevant point is that the main line of debate in spatial justice revolves around the meaning of space and its relationship with society, which brings the concept closer to theoretical discussions of social justice, but distances it from empirical analyses of equity.

Finally, the studies explored here point to strong normative principles about what constitutes a fair distribution of transportation outcomes; however, proposals to change transportation systems have always resulted in an unequal distribution of costs and benefits, and social class strongly influences transportation outcomes. For instance, accessibility levels among white people and the upper classes tend to be far better than among the

middle and lower classes. Among black people, however, social historical trajectories play a major role in explaining accessibility. Focusing solely on distribution is insufficient to address the wide range of demands for justice, since the idea of justice has several interconnected meanings [21]. In this context, transport equity has recently been replaced by the transport justice approach, as an alternative that incorporates other dimensions of justice, i.e., procedural justice and restorative justice concerns, to respond to the needs and rights of all, especially those who have been historically excluded.

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